

Central-Force Problem in Quantum Mechanics

Derivation up to the Radial Wave Equation

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Instructor of Modern Physics, Semester 2026–2

Starting point

We consider the time-independent Schrödinger equation for a particle moving in a central potential:

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}) + U(r) \psi(\mathbf{r}) = E \psi(\mathbf{r}), \quad (1)$$

where the potential depends only on the radial coordinate:

$$U(\mathbf{r}) = U(r).$$

Because the potential is spherically symmetric, it is natural to work in spherical coordinates:

$$\mathbf{r} \equiv (r, \theta, \phi).$$

We now assume a separable form for the wavefunction:

$$\psi(r, \theta, \phi) = R(r) \Theta(\theta) \Phi(\phi). \quad (2)$$

Laplacian in spherical coordinates

In spherical coordinates, the Laplacian can be written as

$$\nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \left(\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \csc^2 \theta \frac{\partial^2}{\partial \phi^2} \right). \quad (3)$$

Substituting $\psi(r, \theta, \phi) = R(r) \Theta(\theta) \Phi(\phi)$ into the Schrödinger equation gives

$$-\frac{\hbar^2}{2m} \nabla^2 (R \Theta \Phi) + U(r) R \Theta \Phi = E R \Theta \Phi. \quad (4)$$

Now expand the Laplacian term explicitly:

$$\nabla^2 (R \Theta \Phi) = \frac{\partial^2}{\partial r^2} (R \Theta \Phi) + \frac{2}{r} \frac{\partial}{\partial r} (R \Theta \Phi) + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} (R \Theta \Phi) + \frac{1}{r^2} \cot \theta \frac{\partial}{\partial \theta} (R \Theta \Phi) + \frac{1}{r^2} \csc^2 \theta \frac{\partial^2}{\partial \phi^2} (R \Theta \Phi). \quad (5)$$

Since R depends only on r , Θ only on θ , and Φ only on ϕ , we obtain

$$\nabla^2 (R \Theta \Phi) = \Theta \Phi \frac{d^2 R}{dr^2} + \frac{2}{r} \Theta \Phi \frac{dR}{dr} + \frac{1}{r^2} R \Phi \frac{d^2 \Theta}{d\theta^2} + \frac{1}{r^2} \cot \theta R \Phi \frac{d\Theta}{d\theta} + \frac{1}{r^2} \csc^2 \theta R \Theta \frac{d^2 \Phi}{d\phi^2}. \quad (6)$$

Therefore, the Schrödinger equation becomes

$$-\frac{\hbar^2}{2m} \left[\Theta \Phi \frac{d^2 R}{dr^2} + \frac{2}{r} \Theta \Phi \frac{dR}{dr} + \frac{1}{r^2} R \Phi \frac{d^2 \Theta}{d\theta^2} + \frac{1}{r^2} \cot \theta R \Phi \frac{d\Theta}{d\theta} + \frac{1}{r^2} \csc^2 \theta R \Theta \frac{d^2 \Phi}{d\phi^2} \right] + U(r) R \Theta \Phi = E R \Theta \Phi. \quad (7)$$

Divide by $R\Theta\Phi$

Dividing both sides by $R\Theta\Phi$, we get

$$-\frac{\hbar^2}{2m} \left[\frac{1}{R} \frac{d^2 R}{dr^2} + \frac{2}{r} \frac{1}{R} \frac{dR}{dr} + \frac{1}{r^2} \frac{1}{\Theta} \frac{d^2 \Theta}{d\theta^2} + \frac{1}{r^2} \cot \theta \frac{1}{\Theta} \frac{d\Theta}{d\theta} + \frac{1}{r^2} \csc^2 \theta \frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} \right] + U(r) = E. \quad (8)$$

Multiply the whole equation by

$$-\frac{2mr^2 \sin^2 \theta}{\hbar^2}.$$

This gives

$$r^2 \sin^2 \theta \left(\frac{1}{R} \frac{d^2 R}{dr^2} + \frac{2}{r} \frac{1}{R} \frac{dR}{dr} \right) + \sin^2 \theta \left(\frac{1}{\Theta} \frac{d^2 \Theta}{d\theta^2} + \cot \theta \frac{1}{\Theta} \frac{d\Theta}{d\theta} \right) + \frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} + \frac{2mr^2 \sin^2 \theta}{\hbar^2} (E - U(r)) = 0. \quad (9)$$

Rearranging, the ϕ -dependent part can be isolated:

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = -\sin^2 \theta \left[\frac{r^2}{R} \left(\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right) + \frac{1}{\Theta} \left(\frac{d^2 \Theta}{d\theta^2} + \cot \theta \frac{d\Theta}{d\theta} \right) + \frac{2mr^2}{\hbar^2} (E - U(r)) \right]. \quad (10)$$

The left-hand side depends only on ϕ , while the right-hand side depends only on r and θ . Therefore, both sides must be equal to a constant. We choose that constant to be $-m_\ell^2$:

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = -m_\ell^2. \quad (11)$$

Thus the ϕ -equation is

$$\frac{d^2 \Phi}{d\phi^2} + m_\ell^2 \Phi = 0. \quad (12)$$

Its general solution is

$$\Phi(\phi) = A e^{im_\ell \phi}. \quad (13)$$

Single-valuedness of the wavefunction requires

$$\Phi(\phi + 2\pi) = \Phi(\phi),$$

which implies

$$e^{im_\ell(\phi+2\pi)} = e^{im_\ell \phi} \implies e^{i2\pi m_\ell} = 1.$$

Therefore,

$$m_\ell = 0, \pm 1, \pm 2, \pm 3, \dots \quad (14)$$

Separate the r - and θ -dependence

Using

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = -m_\ell^2,$$

the previous equation becomes

$$-\sin^2 \theta \left[\frac{r^2}{R} \left(\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right) + \frac{1}{\Theta} \left(\frac{d^2 \Theta}{d\theta^2} + \cot \theta \frac{d\Theta}{d\theta} \right) + \frac{2mr^2}{\hbar^2} (E - U(r)) \right] = -m_\ell^2. \quad (15)$$

Multiply both sides by -1 :

$$\sin^2 \theta \left[\frac{r^2}{R} \left(\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right) + \frac{1}{\Theta} \left(\frac{d^2 \Theta}{d\theta^2} + \cot \theta \frac{d\Theta}{d\theta} \right) + \frac{2mr^2}{\hbar^2} (E - U(r)) \right] = m_\ell^2. \quad (16)$$

Now divide by $\sin^2 \theta$:

$$\frac{r^2}{R} \left(\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right) + \frac{1}{\Theta} \left(\frac{d^2 \Theta}{d\theta^2} + \cot \theta \frac{d\Theta}{d\theta} \right) + \frac{2mr^2}{\hbar^2} (E - U(r)) = m_\ell^2 \csc^2 \theta. \quad (17)$$

Move the θ -dependent term to the right-hand side:

$$\frac{r^2}{R} \left(\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right) + \frac{2mr^2}{\hbar^2} (E - U(r)) = -\frac{1}{\Theta} \left(\frac{d^2 \Theta}{d\theta^2} + \cot \theta \frac{d\Theta}{d\theta} \right) + m_\ell^2 \csc^2 \theta. \quad (18)$$

Now the left-hand side depends only on r , while the right-hand side depends only on θ . Hence both sides must be equal to another separation constant. We choose it to be $\ell(\ell + 1)$:

$$\frac{r^2}{R} \left(\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right) + \frac{2mr^2}{\hbar^2} (E - U(r)) = \ell(\ell + 1), \quad (19)$$

and

$$-\frac{1}{\Theta} \left(\frac{d^2 \Theta}{d\theta^2} + \cot \theta \frac{d\Theta}{d\theta} \right) + m_\ell^2 \csc^2 \theta = \ell(\ell + 1). \quad (20)$$

The polar-angle equation

Multiply the last equation by Θ :

$$-\left(\frac{d^2 \Theta}{d\theta^2} + \cot \theta \frac{d\Theta}{d\theta} \right) + m_\ell^2 \csc^2 \theta \Theta = \ell(\ell + 1) \Theta. \quad (21)$$

Bring all terms to one side:

$$\frac{d^2 \Theta}{d\theta^2} + \cot \theta \frac{d\Theta}{d\theta} + [\ell(\ell + 1) - m_\ell^2 \csc^2 \theta] \Theta = 0. \quad (22)$$

This is the associated Legendre equation. Its solutions are the associated Legendre functions, and regularity on the sphere requires

$$\ell = 0, 1, 2, \dots, \quad m_\ell = -\ell, -\ell + 1, \dots, \ell.$$

The radial equation

Now return to the radial part:

$$\frac{r^2}{R} \left(\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right) + \frac{2mr^2}{\hbar^2} (E - U(r)) = \ell(\ell + 1). \quad (23)$$

Multiply by R :

$$r^2 \left(\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right) + \frac{2mr^2}{\hbar^2} (E - U(r)) R = \ell(\ell + 1) R. \quad (24)$$

Expand the first term:

$$r^2 \frac{d^2 R}{dr^2} + 2r \frac{dR}{dr} + \frac{2mr^2}{\hbar^2} (E - U(r)) R = \ell(\ell + 1) R. \quad (25)$$

Bring the angular-momentum term to the left:

$$r^2 \frac{d^2 R}{dr^2} + 2r \frac{dR}{dr} + \frac{2mr^2}{\hbar^2} (E - U(r)) R - \ell(\ell + 1) R = 0. \quad (26)$$

Now divide by r^2 :

$$\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} + \left[\frac{2m}{\hbar^2} (E - U(r)) - \frac{\ell(\ell + 1)}{r^2} \right] R = 0. \quad (27)$$

This is the radial differential equation in its standard form.

Equivalently, multiplying by $-\hbar^2/(2m)$, we obtain

$$-\frac{\hbar^2}{2m} \left(\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right) + \left[U(r) + \frac{\ell(\ell + 1)\hbar^2}{2mr^2} \right] R = ER. \quad (28)$$

Final result: radial wave equation

Therefore, the radial wave equation for a central potential is

$$\boxed{-\frac{\hbar^2}{2m} \left(\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right) + \left[U(r) + \frac{\ell(\ell + 1)\hbar^2}{2mr^2} \right] R(r) = E R(r)} \quad (29)$$

Optional note: reduced radial function

It is often convenient to define

$$u(r) = rR(r).$$

With this substitution, the radial equation becomes

$$\boxed{-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[U(r) + \frac{\ell(\ell + 1)\hbar^2}{2mr^2} \right] u(r) = E u(r)} \quad (30)$$

which is the one-dimensional Schrödinger-like equation for the reduced radial function.